Pavones, Costa Rica

Golfito's worst El Niño year killed nobody by drought and produced a third of the nation's dengue. The first action is a petition that has worked here before; the cheapest protection is a lid and chlorine; and the question of who actually runs the water is still open.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Puntarenas / Golfito (Climate Impact Lab region containing Pavón district)	No figure published for Costa Rica	Jun 2026 – Feb 2027 (heat only; not applicable here)	Drought → water-service failure → post-drought dengue; transition flood

16/16 recommendations verified. 11 corrected (partially confirmed), 2 confirmed, 1 unverified, 3 contraindicated. One prior-draft finding refuted (Otto deaths were in the north); one Skeptic contraindication refuted (Costa Rica does have larvicide susceptibility data). The funding target moved from $65k to $62k, under a third, so the envelope stands. Exposure figure: Climate Impact Lab, Sep 2026, pre-peer-review, publishes no Costa Rica or Central America figure. Nothing in this report is scaled from it; the historical record carries the hazard instead.

EXECUTIVE SUMMARY — AT A GLANCE



8.39°N 83.14°W, Costa Rica · Natural Earth

- **The risk.** Drought breaks the water service and dengue follows: in 1997–98 Golfito killed nobody by drought but reported 826 of 852 Brunca dengue cases, a third of the national total.
- **First action.** The Alcaldía de Golfito and its emergency committee petition the CNE board to add Golfito to a drought decree by 15 October 2026, the route that worked in 2019.
- **The money.** $35k floor, $62k target, $100k ceiling (single channel). Top placement: $29.7k through ADI Río Claro de Pavones for lids, covers, and chlorine for ~900 households.
- **Do not.** Do not buy land around a spring before the peak: on Costa Rica's most contested coastline a cash buyer reopens land conflicts exactly when the community must trust whoever talks about water.

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* Suggested formulation: check with a health worker before sharing.

EXPOSURE, HONESTLY

WHAT IS AND IS NOT QUANTIFIED

The *451,000 at Risk* release covers heat only and breaks out Africa, Asia, and Brazil. There is no published mortality projection for Costa Rica, and a life-safety framing built on excess heat deaths would not survive contact with anyone who knows the CCSS. What is quantifiable is narrower:

- **Rainfall.** Pacífico Sur ran 25% below normal January to August 2026 (IMN via CNE). IMN forecasts a further ~30% below normal for September to November across the Pacific slope, with El Niño reaching "very strong" November to January. The 40–50% figures circulating are forecasts for the north, not observations for the south.
- **Water systems.** The 2018 Contraloría audit (DFOE-AE-IF-00008-2018) found Las Gemelas de Pavón already on supply two days in five before any drought: 450 people, a 29 m³ tank against 148.5 m³ required, 28 years old. Drought exposes this; it does not create it.
- **Dengue.** Golfito canton reported 826 of 852 Brunca cases in 1998, a third of the national total, and 43.8% seroprevalence in 2005. Brunca led the country in 2024 at 581 per 100,000 at week 26. A population primed for secondary infection is the plausible death mechanism, and it arrives after the rains return.
- **People.** Pavón district, INEC 2022: 7,206 people, 2,556 occupied dwellings, 2.82 per household. No locality-level count exists for Pavones village. Pavón is among the 15 lowest-ranked districts nationally on MIDEPLAN's social development index; the exact rank quoted in the prior draft could not be verified.
- **Status today.** Golfito went from green to yellow on 18–19 September. As of 22 September no ASADA in Golfito or Brunca is on the national rationing list. Rationing here is plausible, not yet on record; every action below is timed to precede it.

WHAT THIS PLACE SHOULD DO

RANKED

Ranked by expected harm averted per unit of effort before the Nov–Jan peak, then by evidence quality after verification. Backing lenses are

* Suggested formulation: check with a health worker before sharing.

shown per action; convergence was checked against shared literature before it counted.

Petition the CNE board to add Golfito to a drought decree, by 15 October

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · ECONOMIST · PHILANTHROPIST | ADAPTATION INVENTORY: CLOSEST PRECEDENT FONDEN (MEXICO)

OWNER	BY WHEN	COST	PREVENTS
Alcaldía de Golfito + Comité Municipal de Emergencias	15 Oct 2026 (2019 took ~6 weeks)	Zero	The 2014 repeat: in drought, outside the fund

Decreto 38642-MP-MAG (2014) named 19 cantons and left out the whole south. The correction from verification is the best precedent in this report: **Golfito was added to the 2019 drought decree, 41944-MP-MAG, by exactly this route**, a CNE Junta Directiva resolution (188-09-19, 4 September) published in La Gaceta on 25 September. Petition the CNE board under Ley 8488 Art. 29 with the IMN Pacífico Sur deficit attached, not only the Presidencia.

Why it ranks first: the World Bank Cat DDO (US$160M) and the Fondo Nacional de Emergencias both gate on the declaratoria, not on the hazard. The Economist's claim that the Cat DDO excludes drought was refuted; its program document has no hazard list and names drought as a priority. The declaration is the whole gate. Inventory precedent: Mexico's FONDEN, an indexed fund whose automatic release reduced deaths (benefit–cost ≥ 3.2), is the closest documented case for release-on-trigger; nothing in the inventory covers Costa Rica.

Point-of-use chlorine and covered storage for every rationed household, before rationing starts*

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE · HISTORIAN · EMERGENCY MGR · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Área Rectora de Salud Golfito + EBAIS + water operators; ADI as fiscal agent	31 Oct 2026, or first rationing decision	~$33/household; $29.7k for 900 hh (est.)	~25% less diarrhoea (low certainty); storage-bred Aedes

Cochrane CD004794 (14 trials, 30,746 participants): household chlorination RR 0.77 (0.65–0.91), **low-quality evidence**, mostly unblinded self-report, larger effect when safe storage is supplied with it. Justify this on illness, not deaths: Costa Rica's diarrhoeal mortality is low, and the $53/DALY figure often quoted for chlorination is a 2007 model for high-child-mortality Africa that does not

* Suggested formulation: check with a health worker before sharing.

transfer.

Procurement correction: NaDCC drinking-water tablets are not on Costa Rican retail shelves (what is sold is pool chlorine). Source Aquatabs through an importer or use liquid hypochlorite with a dosing card. The Skeptic's point stands that lids come off tanks in daily use; pair the lid with the weekly check in the next action. Río Claro de Golfito is in Guaycará district, not Pavón; the Pavón settlements are Pavones, Punta Banco, Río Claro de Pavones, Conte, Zancudo, and Las Gemelas.

Container-cover and yard campaign the week rationing starts, aimed at Feb–Apr 2027

PARTIALLY CONFIRMED | PUBLIC HEALTH · HISTORIAN · FRONTLINE · EMERGENCY MGR | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Área Rectora de Salud Golfito + volunteers; ADI consent in Conte	**Start with rationing; sustain to Apr 2027**	**Volunteer time; larvicide in-kind from MinSal**	**The 1998 and 2024 pattern: post-drought dengue**

The effect size cited by two lenses, OR 0.22 (0.15–0.32), turns out to be **one Cuban observational study** inside the Bowman 2016 meta-analysis, not a pooled estimate; the authors themselves note a "remarkable paucity of reliable evidence." The drought–dengue lag from Lowe 2021 (Brazil) is RR 1.43 at four months overall, but **RR 1.15 in rural areas against 1.60 urban**, and Pavón is rural. The local record (1998, 2005, 2024) is what carries this action, not the imported literature.

Larvicide correction: the Skeptic's "no Costa Rica susceptibility data" was refuted. UCR/MinSal bioassays 2013–16 across eight localities found no temephos resistance (resistance was to adulticides), and the Ministry's vector programme supplies four larvicides (temephos, spinosad, pyriproxyfen, Bti). Request them from the Área Rectora rather than buying them; ask for a Pacífico Sur bioassay in parallel because no southern sample exists and the data are ten years old.

Drill the 2024 national dengue protocol at EBAIS Comte and Alto Conte, with a phone or radio referral fallback

CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CCSS Área de Salud Golfito (8 EBAIS, 32 visit points)	**31 Jan 2027**	**Within CCSS**	**Severe secondary dengue deaths in a 44%-seropositive population**

* Suggested formulation: check with a health worker before sharing.

Nothing to set up. The MinSal/CCSS/PAHO protocol of 19 September 2024 already uses the WHO 2009 warning-sign classification and mandates EBAIS classification and notification at first contact. The gap is operational: EBAIS Comte opens Monday to Friday 8:30–14:30, Alto Conte keeps paper records during internet outages, and Hospital de Golfito is the referral point. Drill it before the case curve, not during it.*

Weekly intake gauging by the fontaneros, a non-meter household census, and a leak-repair kit, from 1 October

PARTIALLY CONFIRMED | EMERGENCY MGR · FRONTLINE · SKEPTIC · PHILANTHROPIST · ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Water operators + ADI volunteers; report to AyA, Municipalidad, CME	1 Oct 2026, weekly thereafter	$5k kit; otherwise volunteer time	Being invisible when allocation happens

Five lenses converged from five rationales, and the convergence survived the shared-bias check because the mechanism is local: Conaflu's own instruction to ASADAs this month is to report supply hours to AyA, the municipality, and the CME. The census reaches what the meter count misses: rented rooms, lodge staff quarters, Ngäbe households. The Contraloría found more than half of AyA's 2023 production unbilled and most ASADAs not metering production at all, so leak loss is plausible but unmeasurable until someone gauges.

Corrected: the prior draft's claim that Golfito "got trucks not wells in 2014–15 because it did not report" has no primary record and is marked unverified. The gauging stands on its own.

Storage tanks and chlorinators for the six Pavón systems through the JUDESUR → Fondo Nacional channel, with a parallel CNE Art. 15 request

PARTIALLY CONFIRMED | ECONOMIST · HISTORIAN · EMERGENCY MGR · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CME Golfito + JUDESUR Junta Directiva + AyA ORAC Brunca	Application filed by 15 Oct, while the ₡3,000M surplus is unallocated	Unfunded pending grant; 2022–23 ran ₡722M for 33 systems	29 m³ tanks serving 450 people at two days in five

* Suggested formulation: check with a health worker before sharing.

The precedent was overstated by two lenses and corrected: the southern ASADA programme of 2022–23 was **₡722–724M of JUDESUR transfers routed through the Fondo Nacional de Emergencias**, not ₡3.12bn and not Art. 15 first-impact money. Thirty-three systems in Osa, Golfito, Buenos Aires, and Coto Brus received tanks, chlorinators, and pipe; no Pavón system is named. That is the proven channel. File the JUDESUR non-reimbursable application now (Ley 9356; five-step technical form to the board), because a diputado has already proposed the surplus for depósito buildings.

Art. 15 of Ley 8488 does let the CNE act "sin que medie declaratoria" for local emergencies, including "materiales para rehabilitar los servicios básicos," but it does not mention slow-onset drought, and a yellow alert does not trigger it by itself. Expediente 25.799, presented 23 September, would add "evolución progresiva"; it is not law. Ask anyway; pair it with the JUDESUR route.

Three 10 m³ tanks ordered now, and a tanker bridge contract that expires on 30 April 2027

PARTIALLY CONFIRMED | ECONOMIST · EMERGENCY MGR · PHILANTHROPIST; SKEPTIC DISSENT RESOLVED | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	TRIGGER
Water operator or ADI, pre-signed	Tanks ordered by 15 Oct (order-only, 6–8 weeks); contract by 30 Nov	~$8k standby + ~$2–3k per tank delivered (unverified)	IMN Río Claro (Golfito) station, 30-day rainfall threshold

* Suggested formulation: check with a health worker before sharing.

The Skeptic was right that Guanacaste 2015 trucking to 38 communities had no end date and became the solution. The resolution is a bridge with a hard expiry, paired with tank enlargement, so the contract cannot outlive the works it is meant to cover. The 10 m³ polyethylene tanks are order-only at the regional distributors with no published price; delivery to Pavones inside eight weeks is a real risk, which is why the order date is October, not December. There is no IMN "Golfito" station; the public automatic station is Río Claro (Golfito), and its data are real-time and uncontrolled.

Plan for the transition rains: pre-stock shelters and agree a Golfo Dulce boat medevac by 1 November

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN | ADAPTATION INVENTORY: PRECEDENT, NICARAGUA BRIDGES (FLOOD-CUT ACCESS)

OWNER	BY WHEN	COST	PREVENTS
CME Golfito + Cruz Roja Golfito; named boat owners, VHF channel	1 Nov 2026 (Otto's precursor floods hit 16 Nov 2016)	Minimal	Isolation as in Nate 2017 at Hospital Tomás Casas

Every death in this canton's record came from water arriving, not leaving. Otto's precursor low dropped up to 300 mm in 24 hours on Golfito, Osa, and Corredores with a red alert and 134 communities affected; Nate cut the Interamericana Sur and isolated the Osa hospital. Two corrections: the ten Otto deaths were in Upala and Bagaces, not the south, and "16 of 72 Brunca EBAIS" was a misread of a national figure. No primary evidence shows the coastal road to Pavones was cut in either event; plan as if it will be, because the first hard rain lands on baked, cleared ground.

Reach Conte and Alto Conte in Ngäbere, through the ADI, before the coffee migration

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH · HISTORIAN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
EBAIS Alto Conte + ADI + a trusted Ngäbere speaker	Before October departures	Volunteer time	The one place with no stream in verano being missed

Conte Burica (11,910 ha, straddling Pavón and Laurel) had 1,863 residents in 2011, 61% indigenous; the "67% Ngäbere" figure is unverified. The MIVAH 2013 diagnostic confirms families migrate for the coffee harvest and return in April and May, and that Alto Conte has no stream nearby in the dry season. Absence is partial, not total: whoever stays is the household with the least water and the least

* Suggested formulation: check with a health worker before sharing.

access. Voice notes in Ngäbere through existing groups, not a poster from the Municipalidad, and nothing without ADI consent.

A SINAC-accredited volunteer fire crew with an equipment cache and one muster point, by 1 January

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN; SKEPTIC PARTLY RIGHT; PUBLIC HEALTH RANKS IT LAST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
SINAC ACOSA + Bomberos Golfito + ADI Bahía Pavón	1 Jan 2027, before the Feb–Apr fire window	~$4k in-kind (unverified)	The 1998 pattern: 1,300 ha in ACOSA, one station across the gulf

The 1998 figures are exact (1,300 ha of a national 64,893 ha). The indigenous brigade precedent was conflated by two lenses: park fires in La Amistad fell from about 200 ha to 10–15 ha, and the 2,000–3,000 ha figure was the 2009–10 territory burn, now "hundreds." Brigades cut both. Accreditation is by decree (SINAC, members 18–60, certified training), so 1 January is tight but feasible if ACOSA moves. Ranked last on evidence: smoke mortality is a ~2% risk increase per 10 μg/m³ and the local source is waste burning, not wildfire.

Post-peak: a second source that is not a stream, starting with a Pavón hydrogeology study

PARTIALLY CONFIRMED · POST-PEAK | HISTORIAN · ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
AyA ORAC Brunca + SENARA; JUDESUR grant candidate	Study commissioned 2027; not a pre-peak action	₡60–150M estimate, unfunded	Surface intakes failing faster than aquifers, with no buffer

* Suggested formulation: check with a health worker before sharing.

The SENARA document cited by the Historian is a 2019 compilation of existing well data for Golfito town, kilometres 3 to 29, about 25 km from Pavón across the gulf. The ex-banana wells yielding 150–365 GPM are real and irrelevant to Pavón. Drilling needs a SENARA permit and a Dirección de Agua concession, so nothing here lands before the peak. JUDESUR has funded an aqueduct before (Volcán), which makes it the payer to approach.

Tariff levers: the ASADA block tariff and the water-source protection tariff

UNVERIFIED · NOT A PRE-PEAK LEVER | ECONOMIST · SKEPTIC | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Water operator assembly + ARESEP	Months, not weeks	Negative net cost in principle	Peak-season demand from lodges and pools going unpriced

The Economist's "voluntary adoption for 2026" is not established. ARESEP's RE-0002-IA-2026 was a partial, temporary suspension of the 2024 block-tariff resolution, and RE-0009-IA-2026 then declared that resolution null. The source-protection tariff exists (Río Celeste was the fourth ASADA granted one, after a year of accompaniment) and funds recharge-area protection rather than land purchase; it is an ordinary tariff process measured in months. Keep it for 2027. The demand-tier agreement with lodges from the prior draft belongs here too, on the Skeptic's Sardinal condition: through the operator's assembly, never State-to-hotel.

* Suggested formulation: check with a health worker before sharing.

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Municipalidad de Golfito	CME Golfito	CNE	Área Rectora de Salud	CCSS Área de Salud (EBAIS)	Water operators	ADIs	Volunteers & boat owners	AyA ORAC Brunca	JUDESUR	Cruz Roja Golfito	SINAC ACOSA & Bomberos	SENARA	ARESEP
1. Petition the CNE to add Golfito to a decree	**A/R**	**R**	**C**											
2. Chlorine and covered storage per household				**A/R**	**R**	**R**	**R**							
3. Container-cover and yard campaign				**A/R**			**C**	**R**						
4. Drill the 2024 dengue protocol					**A/R**									
5. Weekly gauging, household census, leak kit	**I**	**I**				**A/R**	**R**		**I**					
6. JUDESUR tanks and chlorinators, six systems		**A/R**	**C**			**C**			**R**	**R**				
7. Three tanks and an expiring tanker bridge						**A/R**	**R**							
8. Pre-stocked shelters and a boat medevac		**A/R**						**R**			**R**			
9. Reach Conte and Alto Conte in Ngäbere					**A/R**		**R**	**R**						
10. SINAC-accredited volunteer fire crew							**R**	**R**				**A/R**		
11. Post-peak: Pavón hydrogeology study									**A/R**	**C**			**R**	
12. ASADA block and source-protection tariffs						**A/R**								**C**

What the chart shows. **The water operators are accountable for three rows (5, 7, and 12), yet nobody knows who operates the Pavones system.** The funding section names this as its largest uncertainty, so finding the operator comes before any of those rows can start. **The ADIs do the work on five rows (2, 5, 7, 9, and 10).** Both are authorised, but neither has a public record of handling $30–60k, so their capacity is the plan's main dependency. **The two rows that unlock public money sit with national bodies.** Row 1 needs a CNE board resolution and row 6 needs JUDESUR's board, while a diputado has already proposed the surplus for other buildings. Golfito can file, but it cannot decide. **Volunteers carry an R on four rows (3, 8, 9, and 10)**, and only the fire cache, $4k in kind, has money attached.

* Suggested formulation: check with a health worker before sharing.

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal harm averted per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$35k	$62k	$100k (single channel)	Who operates the Pavones water system

Arithmetic. Floor: 900 households × $33 = $29.7k for lids and chlorine, plus $5k gauging and leak kit = $35k; the lens's $30k had omitted the kit. Target: $29.7k + $5k + $18k tanks and bridge + $4k fire cache = $56.7k, plus 10% fiscal-agent overhead = $62k; it moved from $65k, under a third, so the envelope stands. Ceiling: target plus a $20k land-option deposit that is now contraindicated pre-peak and a $15k second tanker window, held at $100k as the limit one ADI can procure and report before December. The 900-household figure is an assumption (35% of the district's 2,556 occupied dwellings) because no locality count exists. **District-wide coverage would be 2,556 × $33 = $84k for placement 1 alone and about $123k in total, which exceeds the single-channel ceiling.** Beyond $100k the balance has to travel through JUDESUR or the CNE, not through the ADI.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
ADI Río Claro de Pavones (fiscal agent); Área Rectora de Salud Golfito supervising	Action 2: lids, covers, chlorine for ~900 households	$29.7k	Pre-positioned stock, direct grant	Figures real but not transferable: Clasen 2007 $53/DALY is an Africa model; Girardot $20.90/household was the incremental cost of insecticide-treated covers in an urban city, ICER $12,097 per 1-point infection cut; Suaya 2007 $313/DALY public perspective. Order-of-magnitude only.	Partially confirmed
Water operators via the ADI	Action 5: gauging and leak-repair kit	$5k	Direct grant	No Central American cost per m³ recovered found. Justified as the information that gates every other placement.	Unverified: operator unknown

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Water operator or ADI, pre-signed	Action 7: three 10 m³ tanks + tanker bridge with expiry	$18k	Pre-signed contract; tanks ordered	$8k standby and $2–3k per tank delivered are estimates; tanks are order-only. Not duplicated: no CNE, AyA, JUDESUR, IMAS, or Cruz Roja line for Pavón found; the "GAM-only" AyA claim was refuted (AyA's plan targets Chorotega and surface systems), and the 2020 57-tank programme was a COVID-era measure, irrelevant here.	Partially confirmed
Bomberos Golfito / SINAC ACOSA	Action 10: fire-crew equipment cache	$4k	In-kind	No cost-per-hectare evidence found.	Unverified
Área Rectora de Salud Golfito	Action 3: larvicide	$0 from grant	Top-up request to MinSal's vector programme	Ministry supplies temephos, spinosad, pyriproxyfen, and Bti; Costa Rican bioassays show no temephos resistance. Buying it would duplicate a public line.	Confirmed as in-kind
Water operator + Rotary / foundation	Spring-land purchase	Dropped pre-peak	—	Contraindicated before the peak; see below. Post-peak via a protection agreement, not a cash buy.	Contraindicated

Recipient check: both ADI Río Claro de Pavones and ADI Bahía Pavón are on Hacienda's DINADECO-authorised list for December 2025 to December 2026, active, not registered in SICOP. The cédula jurídica quoted by the lens could not be verified, and neither ADI has a public record of handling $30–60k. The funder's one thing survives verification: the cheapest protection here is a chlorine dose and a lid, not a tanker, and the only named public money in Golfito this year is a ₡35M generator for the hospital and fire station while ₡3,000M of JUDESUR surplus is being argued over for shop buildings. One percent of that surplus, routed through the proven channel, covers this envelope.

DO NOT DO THIS — CONTRAINDICATED

Advice that circulates for this hazard and would do harm here. Kept even where it overlaps a ranked action, because the wrong version of a right

* Suggested formulation: check with a health worker before sharing.

action is the common failure.

Do not buy land around a spring in Pavones before the peak

`CONTRAINDICATED` `SKEPTIC; HISTORIAN'S PRECEDENT STANDS ELSEWHERE`

The Pilangosta precedent is real: the Hojancha ASADA bought about a hectare at its springs with Rotary and foundation money two years ahead of 2014–15 and coped. But Pavones is Costa Rica's most contested coastline: the 1997 Max Dalton gunfight, the maritime-zone demolitions of May 2025 against seventy-year occupancies, two amparos and a criminal complaint against the mayor still live. A cash buyer at a spring reopens exactly that, in the months when the community needs to trust whoever is talking about water.

Instead: a servidumbre or protection agreement with the landowner, funded post-peak through ARESEP's source-protection tariff or SINAC's payment-for-ecosystem-services scheme, which protect the recharge area without transferring title.

Do not sign tanker trucking without an expiry date

`CONTRAINDICATED AS THE PLAN` `SKEPTIC; RESOLVED INTO ACTION 7`

In 2015 AyA sent two trucks a day to Tamarindo "while the problem is solved definitively," with no end date and, in the reporting, "no clear way out." Trucking that has no expiry becomes the infrastructure. The prior draft's strongest cost argument, El Cairo de Siquirres, where trucking spend reportedly exceeded the cost of an aqueduct over seven years, could not be re-verified this run and is not relied on here.

Instead: a bridge contract on a rain-gauge trigger with a hard expiry of 30 April 2027, paired with tank enlargement and a filed application for permanent works.

Do not frame this as excess deaths, and do not cite forty weeks of dengue forecast skill

`CONTRAINDICATED` `SKEPTIC; PUBLIC HEALTH CORRECTED`

There is no published mortality projection for Costa Rica, and a briefing that borrows a heat number for a drought will lose the argument and the plan with it. Separately, Fuller et al. 2009 does report "up to 40 weeks" of lead from ENSO and vegetation indices, but it is national, in-sample, 2003–07, and never operationally validated; the best cantonal model (Barboza et al. 2023) is validated on a three-month window and its skill decays across it.

* Suggested formulation: check with a health worker before sharing.

Instead: commit a contingent 2027 vector budget in October on ENSO alert status, which strong El Niño years have historically earned, with release triggers at three months. Measure the response by supply hours and the Aedes container index, not by a death toll.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1982–83	El Niño drought; Decretos 13.855-G, 13.916-G, 14.320-G declare Golfito, Osa, Corredores, Coto Brus, Buenos Aires disaster zones (Ley 6832 itself is a tobacco and onion compensation law)	—	The south was on the drought map
1984–85	Banana company withdrawal after the 1984 strike	Golfito district population 17,923 (1963) → ~11,060 (2022)	A duty-free depósito (1990), not water or land
1996	Hurricane César	≥39 dead nationally; Golfito, Osa, Corredores worst hit	—
1997	Pavones land conflict, Max Dalton gunfight	Two killed (prior draft said three)	Litigation for decades
1997–98	Strong El Niño: ACOSA burns 1,300 ha of a national 64,893 ha; Golfito reports 826 of 852 Brunca dengue cases, 33% of the national total	Dengue, not fire or drought, was the harm	1999 cases fell to 77; nothing structural
2005	Dengue seroprevalence survey, Golfito district	43.8% (92 of 210)	A population primed for secondary infection
2014	Decreto 38642-MP-MAG: drought emergency in 19 cantons, none in the south	Five-year vigencia	Guanacaste got wells; the south got nothing on record
Nov 2016	Otto's precursor low: up to 300 mm/24 h in Golfito, Osa, Corredores; red alert; 134 communities	Otto's 10 deaths were in Upala and Bagaces, not here	—

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Oct 2017	Tropical Storm Nate: Interamericana Sur cut, Hospital Tomás Casas (Osa) isolated	Nationally 570,000 without water; 2 Brunca EBAIS closed, 8 partial by 11 Oct	No evidence the coastal road to Pavones was cut
2018	Contraloría audit DFOE-AE-IF-00008-2018: 29 aqueducts, 104 vulnerable communities	Only 7 of 29 met all potability tests; Las Gemelas de Pavón on supply 2 days in 5, 29 m³ tank vs 148.5 m³ required	Score 4.45/10; 2026 follow-up: 32 of 42 ASADAs did no 2025 water tests
Sep 2019	Decreto 41944-MP-MAG adds Golfito to the drought decree on CNE acuerdo 188-09-19	—	**The precedent for action 1**
2022–23	₡722–724M JUDESUR transfers via the Fondo Nacional for 33 southern ASADAs: tanks, chlorinators, pipe; 95% complete; 54,797 people	—	No Pavón system named. The proven channel for action 6
2024	Post-El Niño dengue year	Brunca 581 per 100,000 at week 26, highest in the country	—
May 2025	Municipal maritime-zone demolitions in Pavones: two supermarkets, a bakery, a restaurant, homes, cabinas	Families resident ~70 years	ADI Río Claro led the opposition; amparos and a criminal complaint filed by 28 May
18–22 Sep 2026	CNE alerts: 23 cantons orange, 52 yellow; Golfito yellow. Pacífico Sur −25% Jan–Aug. 32 ASADAs rationing in 74 communities	—	None in Golfito or Brunca on the rationing list

The recurrence nobody has noticed, from the Historian and confirmed by the record: Golfito's worst El Niño year on record killed nobody by drought. It produced a third of the nation's dengue. Rationing fills every barrel in the village, and the mosquito, not the dry tap, is what the failed second wet season delivers here. Budget for October's water and for March's fever.

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WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **Who is the counterparty for water money** versus **who actually operates the system**. The Frontline and Philanthropist lenses route everything through the ADI; the Skeptic says that concentrates a divided town's water decisions in one contested body and wants the ASADA as counterparty. Verification found no registered ASADA for Pavones and instead found that **ADI Bahía Pavón held a 2007 AyA delegation agreement for the Bahía Pavón aqueduct** and was still the convening body in 2019. The operator may be an ADI. One phone call to AyA ORAC Brunca settles this, and it decides two placements.
- **Larvicide only after a susceptibility bioassay** (Skeptic) versus **budget it now** (Public Health, Philanthropist). Resolved toward the latter: Costa Rican bioassays exist and show no temephos resistance, though no Pacífico Sur sample exists and the data are a decade old. The Skeptic's residual point, that a southern bioassay should be requested, is adopted.
- **25% deficit** versus **40–50%**. Both are IMN figures; the first is observed January to August for the south, the second is the forecast for the north. The Skeptic conflated them. The south's outlook is about 30% below normal for September to November.
- **Drought drives dengue here** versus **the rural signal is weak**. Lowe 2021 gives RR 1.15 rural against 1.60 urban, which weakens the imported mechanism for Pavón; the 1998, 2005, and 2024 canton record is what keeps the dengue actions ranked high. What would settle it: whether Golfito's 1998 cases clustered in Golfito town and Río Claro de Golfito, the hotspots the ASIS names, or reached the Pavón settlements.
- **Build a Salitre-style brigade** versus **it will not transfer**. The Skeptic is partly right: Salitre had one cohesive Indigenous Development Association; Pavón's equivalent, ADI Bahía Pavón, is usable and weaker. Resolved into a smaller, accredited crew with a cache.

WHAT NOBODY COVERED

BLIND SPOT

No lens established whether Pavón is rationing today or which of its six systems will fail first; every dengue and chlorination action is timed to a rationing decision nobody has yet recorded. No lens costed the December to March tourism high season, when lodge pools and laundries land on minimum supply; the prior draft had it, the Skeptic's Sardinal warning is the only trace this run. No lens looked at heat for the people who work outdoors here, the fishers, palm and rice labourers, and surf instructors, and no Costa Rica heat figure exists to prompt it. And no lens found a household count for Pavones village itself, so the 900-household placement rests on a stated 35% assumption. Any of these could move the ranking; the first one could void it.

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Each briefing starts from a single, precise place, such as a village, a ward, or a district, because that is the scale at which useful actions differ. We first work out what can honestly be measured there. That usually means the Climate Impact Lab's published figures where they exist, the local rainfall record, census data, and the history of what has actually gone wrong in that place. Where no reliable number exists, we say so plainly rather than stretch a regional figure to fit.

We then put the same question to seven expert perspectives, each run as a separate AI research agent working from real sources. Each one asks what people here can do before the peak to keep each other alive.

- **The Emergency Manager** looks at what failed operationally in past events.
- **The Public Health Scientist** looks at who is most at risk and which interventions have measured effects.
- **The Frontline Responder** looks at how the community actually works and who people trust.
- **The Skeptic** looks for advice that has backfired in comparable places.
- **The Economist** asks who pays, who carries the risk uninsured, and whether any money is pre-arranged to arrive on a trigger.
- **The Historian** looks for the pattern that keeps repeating.
- **The Philanthropist** asks where a grant would do the most good, and how much money the place could realistically absorb before the peak.

The seven briefs are then reconciled. Agreement between lenses counts for less than it appears to, because they often draw on the same famous studies. Where one lens recommends something another calls harmful, both sides are recorded rather than quietly resolved. Every surviving action needs an owner, a date, a cost, and a named payer, or it is marked unfunded. The funding recommendation becomes a bounded range: a floor, a target, and a ceiling, each with its arithmetic shown. Finally, a separate set of agents checks every recommendation against primary sources. They test whether the evidence is real, whether it came from somewhere like this place, and whether the action can be done here in time. Anything found to be harmful is kept in the report under its own heading, because advice that would cost lives is the most important thing a reader can be warned about. The result is a strong, source-checked hypothesis from a simulated panel, not a consensus of the field, and each briefing says so.

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and

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feasibility before the peak by six further agents; the verdict tags carry the result. This is the third run of this skill on Pavones and the first to complete verification; the 24 September draft was published unverified because the search budget ran out, and it is superseded by this document.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only* and publishes nothing for Costa Rica; drought, flood, fire, and dengue are carried here by the historical record rather than by a published number. The Adaptation Inventory was consulted through its data feed (96 entries, 30 countries): it holds no Costa Rica entry and nothing on rural water supply, vector control, or community fire crews. Where a card says "none found," the inventory was checked and was silent.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Pavones, Costa Rica. Climate Impact Lab impact region, as the report records it: Puntarenas / Golfito (Climate Impact Lab region containing Pavón district). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.

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- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted; the closest documented precedent for each action, or its absence, is shown in that action's tags.
- **Health wording.** Passages where a wording error could hurt someone (chlorinating drinking water, and dengue warning signs and referral) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Gaps specific to this run are recorded in the verification banner and in each action's tags.

References (verification status in brackets) · Decreto 38642-MP-MAG, La Gaceta 10 Oct 2014 imprentanacional.go.cr [confirmed] · Decreto 41944-MP-MAG, La Gaceta 25 Sep 2019 imprentanacional.go.cr [confirmed] · Ley 8488 Art. 15 and 29 pgrweb.go.cr [confirmed] · Expediente 25.799 delfino.cr [confirmed, not law] · World Bank Cat DDO II program document worldbank.org [drought exclusion refuted] · CCRIF FAQ ccrif.org [confirmed] · OCHA Central America AA framework unocha.org [confirmed] · Anticipation Hub, Costa Rica anticipation-hub.org [two EAPs, no drought] · INS parametric coffee vidayexito.net [confirmed] · JUDESUR surplus delfino.cr [confirmed] · 2022–23 southern ASADA programme tvsur.co.cr · guananoticias.com [corrected] · ARESEP RE-0002-IA-2026 aresep.go.cr · Río Celeste TPRH aresep.go.cr [partially confirmed] · Cochrane CD004794.pub3 doi.org [confirmed, low certainty] · Clasen 2007 iwaponline.com [not transferable] · Bowman 2016 plos.org [single-study OR] · Lowe 2021 thelancet.com [rural RR 1.15] · Girardot covers pmc.ncbi.nlm.nih.gov [corrected] · Suaya 2007 pubmed.gov [corrected] · Golfito dengue 1998 and 2005 seroprevalence, Lee-Lui et al. 2008 scielo.sa.cr [confirmed] · Brunca 2024 Boletín 26 ministeriodesalud.go.cr [confirmed] · MinSal PMICV larvicide table ministeriodesalud.go.cr · UCR temephos bioassays ucr.ac.cr [confirmed] · Dengue protocol 2024 ministeriodesalud.go.cr [confirmed] · Fuller 2009 iopscience.iop.org · Barboza 2023 plos.org [both partially confirmed] · ASIS Golfito 2015–19 ministeriodesalud.go.cr · INEC 2022 district estimates inec.cr [confirmed] · CGR DFOE-AE-IF-00008-2018 cgr.go.cr [confirmed] · MIVAH 2013 Diagnóstico Golfito Frontera mivah.go.cr [confirmed] · DINADECO authorised list 2025–26 hacienda.go.cr [confirmed] · ADI Bahía Pavón 2007 AyA convenio vlex.co.cr · 2019 meeting tvsur.co.cr · Conaflu rationing list 22 Sep 2026 delfino.cr · CNE alerts 18 Sep delfino.cr · IMN via CNE infobae.com [confirmed] · IMN automatic stations imn.ac.cr · IMN fires 1998 imn.ac.cr · IMN Otto imn.ac.cr [confirmed] · Nate nacion.com · César reliefweb.int · Guanacaste 2015 trucking elmundo.cr [confirmed] · Pilangosta semanariouniversidad.com · Indigenous brigades semanariouniversidad.com [figures corrected] · Pavones 1997 and 2004 ticotimes.net · 2025 demolitions delfino.cr · ticotimes.net [confirmed] · SENARA Golfito hydrogeology senara.go.cr [does not cover Pavón] · Unverified after search: "Pavones out of water four days, July 2015"; Pavón IDS 24.34 / rank 485 of 490; El Cairo de Siquirres trucking cost; Golfito "trucks not wells" 2014–15.

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Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (queried via wp-json/wp/v2/interventions, 96 entries; FONDEN entry; Nicaragua bridges entry)

Generated 25 September 2026 for Pavones, Costa Rica. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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